Henry Bao

+1-437-345-2716 | henry.bao @uwaterloo.ca | https://www.linkedin.com/in/henryhbao | https://github.com/TruVortex

Education

University of Waterloo Waterloo, ON
Bachelor of Computer Science *Sept. 2025 – Apr. 2030*

Technical Skills

Languages: C/C++, Java, Python, JavaScript/Typescript, HTML/CSS, SQL, Golang, Rust, OCaml, Bash
Frameworks: Next.js, React, Node.js, Tailwind, Django, Flask, MediaPipe, Solana, FastAPI, RESTful APIs
Developer Tools: Git, VS Code, PyCharm, CMake, Linux, Docker, Claude Code, Cursor, Terraform, Azure, Excel
Libraries: TensorFlow, Keras, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, ffmpeg, PostgreSQL

Experience

Infrastructure Support Specialist May 2026 – Now
Teranet *Toronto, ON*

- Acted as the primary responder for alerts and incidents from Spectrum, Solarwinds, CommVault, or ad-hoc events
- Executed time-sensitive patching tasks during planned maintenance windows with zero unexpected downtime
- Fulfilled change requests for server builds and decommissions across various hardware, host (on-prem and Azure), and operating system configurations, allowing for system provisioning and hardening
- Managed on-site vendor relations, supervising contractors and ensuring adherence to facility security protocols
- Streamlined deployments and automated tasks using the infrastructure as code (IaC) tools Terraform and Ansible

Projects

Truv Compression | *Rust, Algorithms* Jun. 2026

- Built a DEFLATE-inspired lossless compression utility combining LZ77 compression and dynamic Huffman coding
- Implemented parallel block compression and decompression using rayon, partitioning files into independently compressed blocks to exploit multicore execution
- Optimized LZ77 match finding using SWAR (SIMD Within A Register) techniques, replacing byte-by-byte comparisons with 64-bit word comparisons and bit-scanning operations
- Built bit-level compression primitives including Huffman tree serialization for compact binary encoding

Tiny Compiler | *OCaml, Dune* Jun. 2026

- Designed and implemented an end-to-end compiler for a functional language, supporting lexical scoping, first-class functions, conditionals, and let-bindings
- Implemented Hindley–Milner type inference (Algorithm W) with unification, polymorphic let-generalization, instantiation, and occurs-check detection for fully inferred static typing
- Developed compiler transformation passes including closure conversion, intermediate representation generation, constant propagation, constant folding, beta reduction, and dead branch elimination
- Built a custom stack-based virtual machine and bytecode instruction set supporting lexical closures, function calls, environment capture, branching, and runtime execution

Tiny Binary Serializer | *OCaml, Dune, Bigstringaf* May 2026

- Built a type-safe binary serializer using GADTs to encode runtime schemas with compile-time type guarantees
- Included support for product types, sum types, optional values, strings, and zero-copy buffer slices
- Implemented variable-length integer encoding and decoding for compact representations of serialized metadata

p2p-file-sync | *Golang, Networking, GitHub Copilot* Feb. 2026

- Built a decentralized file sync engine, utilizing Merkle Trees to achieve $O(\log n)$ state reconciliation across peers
- Optimized bandwidth and memory efficiency by implementing content-addressable storage and SHA-256 chunking
- Created a delta sync protocol using rolling hashes and copy/literal operations to transfer only modified file regions
- Engineered a zero-trust network layer over QUIC, using STUN-based UDP hole punching for NAT traversal

Achievements

CEMC Contests: 14 honour rolls in mathematics and computing contests including one first-place and one runner-up
MAA Contests: 3 time qualification to the selective AIME mathematics contest
Advanced Coursework: 2x 4.0 and 4x 3.9 in advanced math and computer science courses
Certifications: Microsoft Certified: Azure Fundamentals (AZ-900) and HashiCorp Certified: Terraform Associate (004)